TCD Technical Customer Documentation

Preliminary

This preliminary TCD documents the present status of the agreed specifications. It will be confirmed when all validation has been completed with positive results.

Preliminary



Product designation:	Camera Head 3MP SVC
Version:	1.0
Date:	2024-05-13
Customer:	Volvo Truck

Table of contents

1 Product identification 4

2 General product description 5

2.1 Main functions and characteristics of the product 5
2.2 Intended use 6
2.3 Product safety 7
2.3.1 Functional safety 7
2.3.2 Data protection, Cybersecurity aspects 8
2.3.3 Safety and warning notes 9
2.4 Labeling of the product 10
2.5 Dimensions and weights 11
2.6 Power consumption / power output 12
2.7 General remarks on service, repair, and maintenance 12
2.8 Information on disposal and recycling 12

3 System description 12

3.1 System of Interest (SOI) 12
3.1.1 Software Interfaces 13
3.1.2 Hardware Interfaces 14
3.1.2.1 Optical Interface 14
3.1.2.2 Electrical Interface 16
3.1.3 Vehicle Integration Interfaces 21
3.1.3.1 Mounting Interface 21
3.1.3.2 General recommendations bracket design 22
3.1.3.3 Galvanic Isolation from Vehicle Ground 23
3.1.3.4 Poka Yoke and Camera Orientation 23
3.1.3.5 Camera Head with Pre-mounted Bracket 23
3.2 Product interface 24
3.2.1 Mechanical interface of connector plug 24
3.2.2 Electrical interface of connector plug 25
3.2.3 Wiring Harness 25

4 Technical data with measured variables and measuring conditions 26

4.1 Functions, function states (modes of operation), functional characteristics and boundary conditions 26
4.2 Mechanical characteristics 26

Preliminary

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4.3 Electrical characteristics 27
4.4 Climatic characteristics 27
4.5 Chemical characteristics 27
4.6 Acoustic characteristics 27
4.7 Lifetime 27
4.8 Transport, assembly, start and end of operation, storage 28
4.8.1 Transport 28
4.8.1.1 Packaging of Camera Head in RB plant 28
4.8.1.2 Packaging and shipping of preassembled components from Tier1 to OEM 29
4.8.1.3 Transport conditions 29
4.8.2 Assembly 29
4.8.2.1 Unpacking and Typical Assembly 29
4.8.2.2 Camera Orientation 30
4.8.2.3 Forces on Camera Head 31
4.8.2.4 Dropped Devices 31
4.8.2.5 ESD Protection 31
4.8.2.6 Coax Connector Assembly Hints 32
4.8.2.7 Continuity Test 32
4.8.3 Start and End of Operation 33
4.8.3.1 Start of Operation 33
4.8.3.2 End of Operation 33
4.8.4 Storage 33
5 Series-accompanying tests 34
6 Testing 35
6.1 Testing by Bosch 35
6.1.1 Climatic-dynamic Validation 35
6.1.2 Electrical Validation 35
6.1.3 EMC / ESD Validation 35
6.2 Testing by the customer 36
6.2.1 Testing by the customer 36
7 Assessment of products returned from 0-mileage and from the field 37
8 Appendices and references 38
9 History 39
10 Contact person 40

Preliminary

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1 Product identification

Product designation:	Camera Head 3MP SVC
Type designation:	Surround View Camera (SVC)
Product part number:	To be defined
Development part number:	B2-Sample: Camera assembly (w/ bracket): 0275B02093-01 Single camera (w/o bracket): 0275B02094-01 C1-Sample: Camera assembly (w/bracket): 0275B02127-01 Single camera (w/o bracket): 0275B02128-01
Offer drawing number:	Camera assembly (w/ bracket): B2-Sample: 0273A02227 C1-Sample: 0273A02229 Single camera (w/o bracket) B2-Sample: 0273A02234 C1-Sample: to be defined
Name of customer:	Volvo Truck
Customer specification:	To be defined
Number:	
Edition/version:	
Date:	
Title:	
Application:	Support ADAS function
Further applicable documents:	
Volvo specification drawing	78795952
[Document 2]	
[Document ...]	

The product has been developed, validated, and released exclusively for use in these applications:
Automotive surround view camera with vehicle mounting location on the exterior/ in cavity, open towards exterior.

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2 General product description

2.1 Main functions and characteristics of the product

Note: Camera Heads assembly w/ bracket is not considered in this preliminary TCD. The description will be followed when a common agreement of design and handling is established.

The Camera Head transforms a continuous optical signal of a scene into a sequence of electrical image signals and transmits this image data to the connected Electronic Control Unit (ECU) by a coaxial cable.
The Camera Head is configurable by the ECU via Inter-Integrated Circuit (I2C) bus.

More detailed information regarding functionality and characteristics can be found in the following chapters:

- Optical characteristics: Please refer to chapter "3.1.2.1 Optical Interface"
- Electrical characteristics: Please refer to chapter "3.1.2.2 Electrical Interface"
- Product interface (connector) characteristics: Please refer to chapter "3.2 Product Interface"
- Mechanical characteristics, climate characteristics, chemical characteristics, and function description: Please refer to chapter "4 Technical Data with measured variables and measuring conditions"

The coordinate system of the Camera Head, illustrated in **Fehler! Verweisquelle konnte nicht gefunden werden.**, is a right-handed Cartesian coordinate system which is defined as follows (according to ISO 23150):

- The origin is defined as the centre of the image sensor pixel array.
- The z-axis is defined in direction of the columns of the image sensor (nominal position).
- The y-axis is defined in direction of the rows of the image sensor (nominal position).
- The x-axis is defined in direction of the optical axis of the Camera Head with positive direction from the image sensor towards the lens.

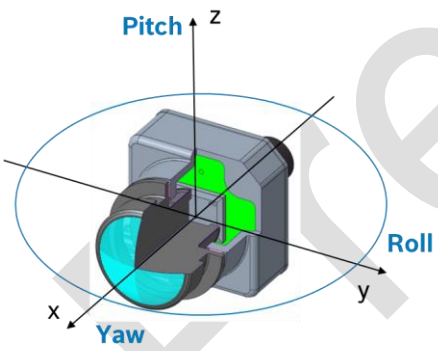


Figure 1: Coordinate system of the Camera Head (according to ISO 23150)

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2.2 Intended use

The product is released based on the regulatory requirements directly applicable to the product at the time of TCD creation in the following target market(s):
Afghanistan (AF), Albania (AL), Algeria (DZ), American Samoa (AS), Andorra (AD), Angola (AO), Anguilla (AI), Antigua and Barbuda (AG), Argentina (AR), Armenia (AM), Aruba (AW), Australia (AU), Austria (AT), Azerbaijan (AZ), Bahamas (BS), Bahrain (BH), Bangladesh (BD), Barbados (BB), Belgium (BE), Belize (BZ), Benin (BJ), Bermuda (BM), Bhutan (BT), Bolivia (Plurinational State of) (BO), Bonaire, Sint Eustatius and Saba (BQ), Bosnia and Herzegovina (BA), Botswana (BW), Brazil (BR), Brunei Darussalam (BN), Bulgaria (BG), Burkina Faso (BF), Burundi (BI), Cabo Verde (CV), Cambodia (KH), Cameroon (CM), Canada (CA), Cayman Islands (KY), Central African Republic (CF), Chad (TD), Chile (CL), China (CN), Hong Kong (HK), Macao (MO), Colombia (CO), Comoros (KM), Congo (CG), Cook Islands (CK), Costa Rica (CR), Côte d'Ivoire CI), Croatia (HR), Cuba (CU), Curaçao (CW), Cyprus (CY), Czechia (CZ), Congo, Democratic Republic of the (CD), Denmark (DK), Djibouti (DJ), Dominica (DM), Dominican Republic (DO), Ecuador (EC), Egypt (EG), El Salvador (SV), Equatorial Guinea (GQ), Eritrea (ER), Estonia (EE), Eswatini (SZ), Ethiopia (ET), Falkland Islands (Malvinas) (FK), Faroe Islands (FO), Fiji (FJ), Finland (FI), France (FR), French Guiana (GF), French Polynesia (PF), Gabon (GA), Gambia (GM), Georgia (GE), Germany (DE), Ghana (GH), Gibraltar (GI), Greece (GR), Greenland (GL), Grenada (GD), Guadeloupe (GP), Guam (GU), Guatemala (GT), Guernsey (GG), Guinea (GN), Guinea-Bissau (GW), Guyana (GY), Haiti (HT), Honduras (HN), Hungary (HU), Iceland (IS), India (IN), Indonesia (ID), Iraq (IQ), Ireland (IE), Isle of Man (IM), Israel (IL), Italy (IT), Jamaica (JM), Japan (JP), Jersey (JE), Jordan (JO), Kazakhstan (KZ), Kenya (KE), Kiribati (KI), Korea, Republic of (KR), Kosovo (XK), Kuwait (KW), Kyrgyzstan (KG), Lao People's Democratic Republic (LA), Latvia (LV), Lebanon (LB), Lesotho (LS), Liberia (LR), Libya (LY), Liechtenstein (LI), Lithuania (LT), Luxembourg (LU), Madagascar (MG), Malawi (MW), Malaysia (MY), Maldives (MV), Mali (ML), Malta (MT), Marshall Islands (MH), Martinique (MQ), Mauritania (MR), Mauritius (MU), Mayotte (YT), Mexico (MX), Micronesia (Federated States of) (FM), Monaco (MC), Mongolia (MN), Montenegro (ME), Montserrat (MS), Morocco (MA), Mozambique (MZ), Myanmar (MM), Namibia (NA), Nauru (NR), Nepal (NP), Netherlands (NL), New Caledonia (NC), New Zealand (NZ), Nicaragua (NI), Niger (NE), Nigeria (NG), Niue (NU), North Macedonia (MK), Norway (NO), Oman (OM), Pakistan (PK), Palau (PW), Palestine, State of (PS), Panama (PA), Papua New Guinea (PG), Paraguay (PY), Peru (PE), Philippines (PH), Poland (PL), Portugal (PT), Puerto Rico (PR), Qatar (QA), Moldova, Republic of (MD), Réunion (RE), Romania (RO), Rwanda (RW), Saint Barthélemy (BL), Saint Kitts and Nevis (KN), Saint Lucia (LC), Saint Martin (French Part) (MF), Saint Vincent and the Grenadines (VC), Samoa (WS), San Marino (SM), Sao Tome and Principe (ST), Saudi Arabia (SA), Senegal (SN), Serbia (RS), Seychelles (SC), Sierra Leone (SL), Singapore (SG), Sint Maarten (Dutch Part) (SX), Slovakia (SK), Slovenia (SI), Solomon Islands (SB), Somalia (SO), South Africa (ZA), South Sudan (SS), Spain (ES), Sri Lanka (LK), Sudan (SD), Suriname (SR), Sweden (SE), Switzerland (CH), Taiwan, Province of China (TW), Tajikistan (TJ), Tanzania, United Republic of (TZ), Thailand (TH), Timor-Leste (TL), Togo (TG), Tokelau (TK), Tonga (TO), Trinidad and Tobago (TT), Tunisia (TN), Türkiye (TR), Turkmenistan (TM), Turks and Caicos Islands (TCA), Tuvalu (TV), Uganda (UG), Ukraine (UA*), United Arab Emirates (AE), United Kingdom of Great Britain and Northern Ireland (GB), United States of America (US), Uruguay (UY), Uzbekistan (UZ), Vanuatu (VU), Holy See (VA), Venezuela (Bolivarian Republic of) (VE), Viet Nam (VN), Virgin Islands (British) (VG), Virgin Islands (U.S.) (VI), Yemen (YE), Zambia (ZM), Zimbabwe (ZW).

(*) UA excluding RU occupied territories.

Note: Country list to be aligned between Bosch and the customer. The final target market list must be agreed during the development project.

Deliveries and services (fulfilment of contract) shall be subject to the provision that there are no obstacles to performance due to national or international (re-)export control regulation, in particular embargoes or other sanctions.

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The product can be used in a released target market in the applications set out in chapter 1, subject to the limits, conditions and other specifications described in this TCD ("Intended use").

In the case that the customer wants to use the product outside the Intended Use, the customer shall

(1) on its own responsibility evaluate and comply with any regulatory requirements resulting therefrom for the product and evaluate and ensure the usability of the product for the area of application intended by the customer or

(2) obtain a new, extending Bosch release, which shall be ordered separately from Bosch (e.g. by means of a change request).

The Camera Head is not intended to work stand-alone. It is designed as a component working in conjunction with a dedicated ECU as a system with intended use in an automotive vehicle with defined environmental loads and function profiles. These loads and profiles are agreed between the vehicle manufacturer and the Robert Bosch GmbH at the beginning of a development project or product adaptation project. Details and scope of validation tests are specified in test catalogues. During validation tests the products have to withstand the specified or equivalent loads.
All changes to the environmental loads or usage profiles for the Camera Head shall be indicated to and released by the Robert Bosch GmbH.

The Camera Head is released for agreed installation positions and viewing orientations for specific OEM vehicles.
If requested by the OEM, additional applications can be released after start of production. This includes new positions and orientations of the Camera Head.

The test modes of the serializer and imager are not assessed and released by Bosch. The correct usage of these test modes is in responsibility of the controlling ECU.

In order to prevent misuse of the camera, please consider the safety and warning notes given in chapter "2.3.3 Safety and Warning Notes**"** on Page 9.

2.3 Product safety

2.3.1 Functional safety

Component with ASIL rating

Bosch points out that the ASIL-classified requirements as per ISO 26262, their implementation and the assumptions made for this purpose are documented in the following documents:

- Development Interface Agreement (DIA)
- Safety case
- Assumptions of use / Requirements to external systems

The customer has to ensure that the component is suitable to be used within the overall system regarding any functional safety requirements assigned to the component in the overall system.

Information on interpretation of ASIL B(D): Bosch assumes the ASIL decomposition is made between the Bosch delivery scope and another sensor technology (e.g. Radar, Lidar, ...), such that there exists heterogeneous redundancy regarding all safety requirements with ASIL B(D). I.e., it is sufficient for Bosch to follow an ASIL B capable development process for all parts of the Bosch delivery scope, as they can be no initiators for common cause failures.

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To enable the integration into a system/function that has ASIL D rating on vehicle level by the customer, Bosch can provide additional analyses upon request. Those requests must be documented in the DIA.

⚠CAUTION

Exemption clause for current project phase (B-Sample status):
The product is not released for series production yet, i.e. it does neither fully comply with the requirements of ISO 26262 nor with the requirements of applicable QM standards. The customer has to ensure that the capabilities of the Bosch deliveries are sufficient for the intended and foreseeable use and that there is no unreasonable safety risks.

2.3.2 Data protection, Cybersecurity aspects

Note: Detailed scope and requirements for cyber security are not available yet, which shall be analyzed and aligned between Bosch and the customer. Bosch determines the conformity to the State-of-the-Art of Cybersecurity of its products on a regular basis. If the state of the art or the applicable laws change before end of delivery of the product, Bosch reserves the right to propose the necessary changes to the product via the change request process.

General information about cybersecurity
The product was developed within an ISO/SAE 21434 certified Cyber Security Management System and corresponding cybersecurity activities were done.

Cybersecurity Claims
The product contains functions which could processes personal data. If so, the product retains the risk of discloser of person identifiable information. As the data controller, the customer designs the processing in detail. It is expected that the customer evaluates if the protective measures of the product are sufficient for the dedicated use case.

Irrespective of the high quality of the delivered component, the product retains the risk that currently unknown cybersecurity vulnerabilities get discovered in the future. As risk mitigation Bosch has installed a Cybersecurity monitoring and further continual cybersecurity activities according to ISO/SAE 21434. In case a cybersecurity vulnerability in a Bosch product is discovered or disclosed by or to the customer it is highly recommended to inform Bosch up-front its publication via PSIRT@bosch.com.

The product contains components which are only partly resistant to fault injection and side channel attacks. Means the product retains the risk of the extraction of sensitive data (e.g., Keys, Intellectual Property, SW) via physical attacks. In consequence, it is expected that no strict confidential data is stored which could potentially lead to a high damage impact.

The product retains the risk that intellectual property (“IP”) from Bosch, the customer and in some cases 3rd parties can be disclosed during the software distribution. For the purpose of IP protection and avoidance of re-engineering, it is expected that delivered software is handled as confidential during the distribution process to the customer, into plants, workshops and Firmware Over-the-Air (FOTA) updates. It is expected that the customer has implemented appropriate measures within his Customer’s Software Logistic Infrastructure (IT network) and associated processes to secure and restrict the access to delivered software.

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The product retains the risk of discloser of confidential data (including but not limited to personal data) during the decommissioning process of the customer or 3rd parties. It is expected that the customer evaluates and adequately mitigates the risk on vehicle level or within the logistic infrastructure.

The product retains the risk of data manipulation in his memory. It is expected that the interfaces, which are designed or implemented by the customer or 3rd parties, do not offer any mechanism to alter the memory content, besides the regular software update (reflash) and intended changes by documented diagnostic services (e.g., UDS).

It is further expected that interfaces, which are designed or implemented by the customer or 3rd parties, and which are capable to change the memory content, are well secured and limited to engineering purposes (e.g., XCP).

The product retains the risk of discloser of confidential data by third-party software, if implemented. It is expected that the interfaces designed or implemented by the customer or 3rd parties do not read confidential data, cryptographic material or software with intellectual property, unless Bosch expressly permits this.

The product retains the risk that software, which is designed or implemented by the customer or 3rd parties, could compromise the intended function of the product. It is expected that the customer validates the operational behavior.

The product retains the risk of a Denial-of-Service (DoS) attack via vehicle internal communication interfaces, because there is currently no known effective mechanism within individual E/E components available. It is expected that the customer evaluates and adequately mitigates the risk on vehicle level.

The product retains an elevated risk regarding the system (-function) availability of the product, caused by the limited effectiveness of the cybersecurity measures implemented in the diagnostic interface, services and routines. It is expected that the customer has evaluated potential "availability" risks (e.g., caused by malicious diagnostic requests) with the scope of the overall system and have implemented adequate cybersecurity measures in the vehicle E/E-architecture (Further information and examples of potential critical diagnostic services can be found e.g., in SAE J 3138).

2.3.3 Safety and warning notes

⚠WARNING

In order to keep the camera head fully functional over the guaranteed lifetime, following advises shall be strongly considered:

- The camera head is only designed for performing the video functions described in this TCD. The camera head is neither recording the video data, nor does it display the data.
- The camera head is only designed for being handled by trained staff.
- The camera head is only designed for use in approved vehicles.
- The camera head is not designed for use without being firmly assembled in a bracket described in this TCD.
- The camera head does not have control elements.
- The camera head could exhibit hot surfaces in extreme weather conditions
- The camera head is not to be opened. It is delivered ready to use.
- No modifications to the camera head or the camera head housing are allowed. This includes removing of parts as well as adding of parts. The camera head is to be used as delivered.

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- After unboxing, the camera head has to be mounted into the vehicle's mounting bracket immediately. It must not be stored or operated when it is not correctly assembled in the vehicle. Long−term storage conditions apply as long as the camera head is in shipping state.
- Writing of data into Imager OTP is prohibited. Consequence would be a functional safety error flag and hence depending on the defined safe state possibly a non-functional camera.
- In order to comply with UN/ECE R46 and UN/ECE R26 the camera is not intended to stick out of the beauty cover by more than 5mm.

⚠WARNING

Misaligned camera head in the vehicle can cause improper system operation. The camera head must be calibrated by an authorized dealer whenever it was removed from or strongly displaced within its bracket, e.g. in case of camera head or windshield replacement.

Besides the warning notes above, please consider the recommendations regarding transport and assembly described in chapter "4.8 Transport, Assembly, Start and End of Operation, Storage".

2.4 Labeling of the product

The Camera Head is labelled on those sides of the camera housing, which are not intended as reference sides for mounting into the vehicle bracket.

The label contains the following data:

- Volvo Part Number (Single Camera)
- Volvo Part Number ID (Single Camera)
- Volvo Trademark
- Volvo Part Number (Assembly)**
- Volvo Part Number ID (Assembly)**
- E1 Marking
- ECE10 Code
- Datamatrix Code ECC200
- Country of Origin
- BOSCH Symbol
- Bosch Part Number

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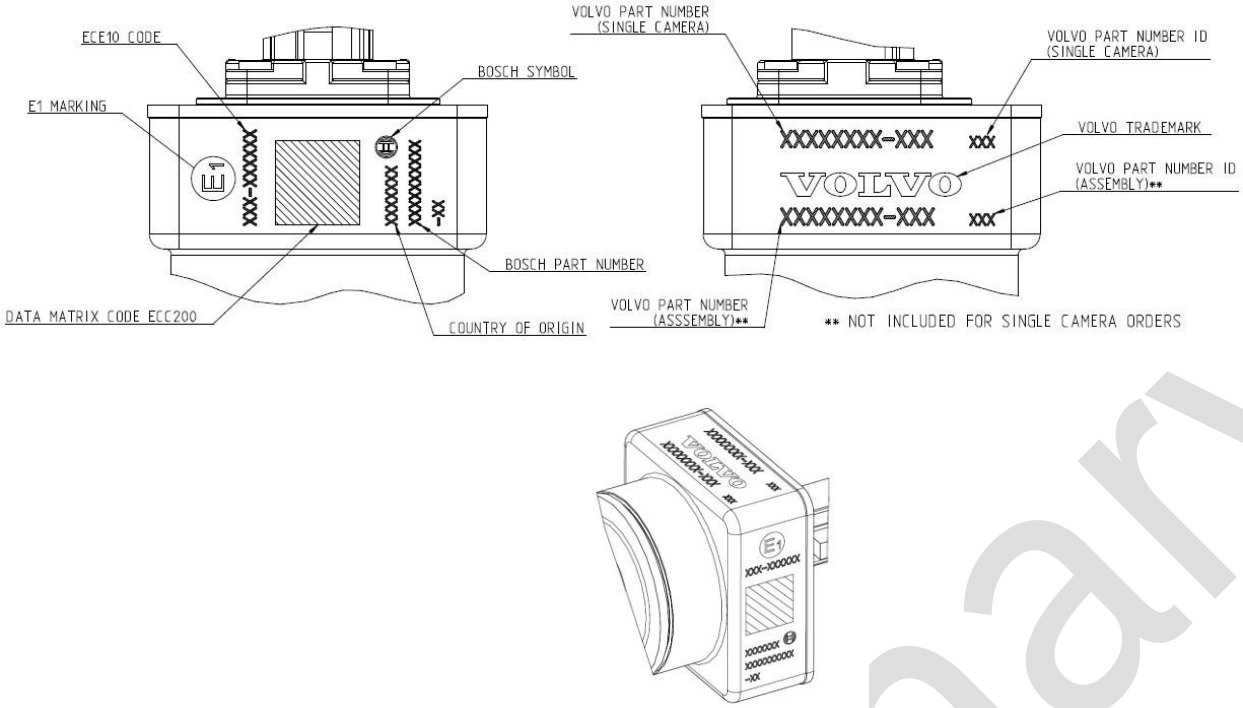


Figure 2: Label of camera head.

Final design freeze and release of the customer label typically with C-Sample. The exact content of the label is described in the offer drawing. Usage of additional labels has to be discussed with Robert Bosch GmbH.

2.5 Dimensions and weights

The box dimensions of the Camera Head are shown in Figure 3. All shown dimensions are without tolerances.

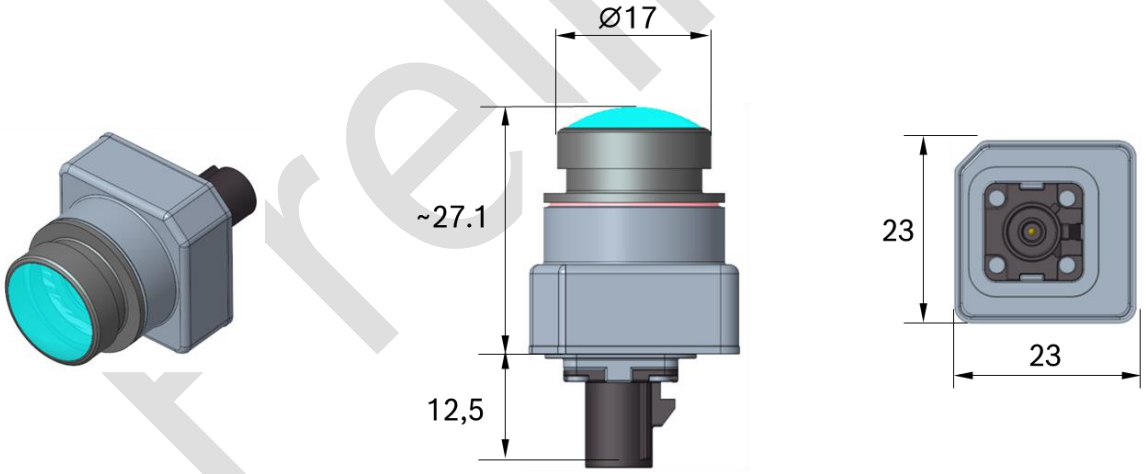


Figure 3: Box dimensions of Camera Head

The approximate weight of the Camera Head is <30g.
The approximate weight of the Camera Head with pre mounted bracket (pre Project Change Request-PCR03) is <50g. The weight including the PCR03/PCR04 needs to be defined.

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2.6 Power consumption / power output

The Camera Head is targeted for a power consumption <1.3W (60fps, nominal voltage (12V)). The power consumption will be increased in case the lens heating function is activated (max. total power consumption <5W).

2.7 General remarks on service, repair, and maintenance

Repair of the product is not possible.

Reason: Due to the camera head's housing with welding interface, it cannot be opened without destruction. Additionally, during repair the lens could be de-aligned and particles could enter the optical path.

Impurities and scratches on or damage of the surface of the front lens of the Camera Head can lead to a degradation of the image quality. Therefore, it must be avoided to load mechanical stress especially on the front lens or to touch the front lens.

If a cleaning of the lens surface is necessary for the Camera Head with a mounting location outside of the vehicle and usage without a cover glass, the following recommendations have to be considered:

- It is recommended to maintain a minimum distance of 15 cm to the Camera Head's main lens when using a jet cleaner (max. 8000 kPa, 80° C).
- Cleaning must be carried out wet. Coarse impurities (visible cover with dirt) must be removed by rinsing. Bright stripes can be removed with a clean cotton cloth and water.
- In addition, it is recommended to avoid treating the front lens during a polishing process.

2.8 Information on disposal and recycling

Based on size of camera head and bracket installation a separate recycling concept for the camera head is not applicable and therefore not intended. No dismantling information will be provided.
The camera head will stay in installation area.

3 System description

3.1 System of Interest (SOI)

The Camera Head is part of an Advanced Driver Assistance System (ADAS) system, consisting of a central ECU and one or more Camera Heads. The complete system is not in scope of this TCD and has to be described in a separate document. This document covers only the Camera Head.
The Camera Head as part of the superordinate system and the functional view of the image chain can be seen in Figure 4.

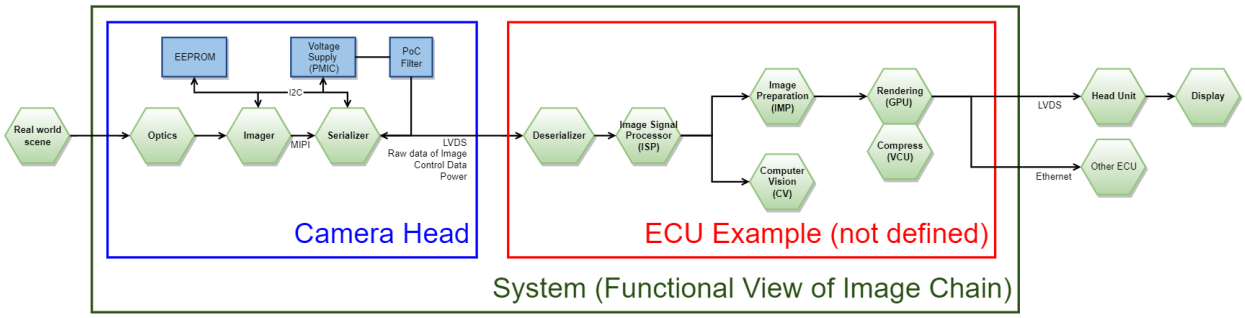


Figure 4: Camera Head within the superordinate system

The optical signal of the environment around the car is projected onto the image sensor by the lens, where it is transformed into an electrical signal and transmitted via the serializer to the ECU.

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The Camera Head is not connected directly to vehicle network but via a coax cable and connector to the ECU. The ECU protects the camera from vehicle network noise. The ECU controls the entire camera functionality including power up, configuration, video streaming and diagnosis features via the back channel.
A more detailed block diagram of the camera head architecture can be seen in **Fehler! Verweisquelle konnte nicht gefunden werden.**

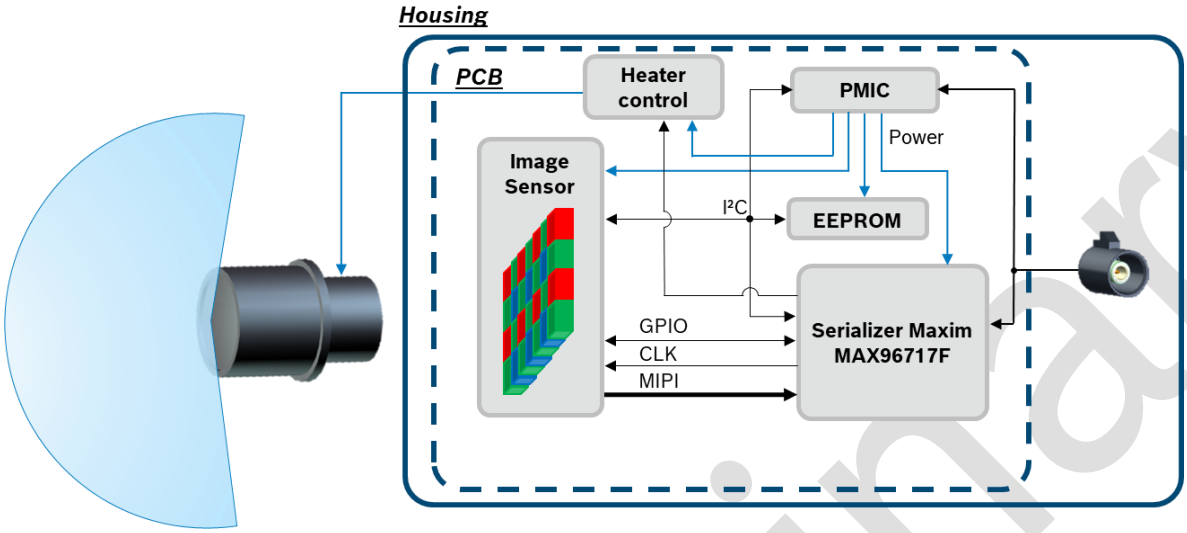


Figure 5: Detailed block diagram of Camera Head Architecture (illustration w/o lens heating functionality).

The Camera Head's package concept relies on a one-PCB-solution and the general product interface for the customer is a coax connector.
The input path of the PCB consists of a PoC filter, which isolates between power and high speed data signals:

- The power is further distributed among the different components by the Power Management Integrated Circuit (PMIC).
- The high speed signals are transmitted by the Serializer which maintains the communication interface to the ECU via video data (forward channel) and control data (back channel).

The Camera Head exhibits a non-volatile memory (EEPROM) for storage of calibration & label data, security signatures and further customer specific data.

Additionally, the lens is equipped with a heating functionality, in order to remove thin ice layers on the front lens. The power for lens heating is switched on and off via a separate control element. Enabling of the control element is done by ECU via the serializer inside of the camera head.

The main interfaces of the Camera Head in the system context are the following:

- Hardware interface (refer to chapter 3.1.2)
  - Optical interface (refer to chapter 3.1.2.1)
  - Electrical interface to the ECU (refer to chapter 3.1.2.2)
- Vehicle integration interfaces (refer to chapter 3.1.3)
- Product interface (related to the coax connector of the camera head, refer to chapter 3.2).

3.1.1 Software Interfaces

The Camera Head exhibits no software and hence, has no software interface. The Camera Head internal configuration is written via I2C by the ECU.

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3.1.2 Hardware Interfaces

3.1.2.1 Optical Interface

3.1.2.1.1 General Optical Characteristics

The main characteristics of the optical interface of the camera head are shown in Table 1.

Table 1: Optical interface characteristics.

Parameter	Value	Unit	Notes
Optical interface			
Horizontal FoV	200°	°	Nominal value on camera level ROI = 1920 x 1536
Vertical FoV	165°	°	Nominal value on camera level ROI = 1920 x 1536
Focus distance	2	m	complies with required sharpness corridor from 1m to 25m
Sharpness (center)	TBD		Based on requirement agreement (open)
F-number	2.0		Design value
Relative illumination	50% @ 75° OA		OA = Object angle Value on Camera Level
Spectral range	410 ... 680	nm	Design Value
Resolution	1920 x 1536	Px	Total pixel array, which can be read-out. Effective resolution might be chosen smaller to cover intrinsic tolerances (camera level) and extrinsic tolerances (vehicle level)
Pixel size	3	µm	Design value
Pixel technology	Split pixel		Design
Color Filter Array	RGGB		Design

3.1.2.1.2 Straylight Cover

In case a stray light cover is used with the Camera Head, its geometry, material and surface roughness should be designed in a way such that no parts are blocking the FOV of the Camera Head and no stray light is introduced by the part.

3.1.2.1.3 Cover Glasses

In case, the Camera Head is used in conjunction with a cover glass or behind a windshield the following influences have to be considered:

- **Stray light effects:** For example, ghosts due to back reflections and internal reflections in the cover glass, halos and beams due to scratches and imperfections in/on the cover glass, etc.
- **Transmission losses:** The transmission of glass and plastic is reduced the higher the incident angle on a surface is. Additionally, the transmission is influenced by the material absorption and coatings.

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- **Change of distortion and reduction of sharpness:** In case a ray does not pass an optical surface perpendicularly, the ray will be refracted. The refraction will lead to a change of distortion, which can depend on wavelength, and can also lead to a defocus of the camera image. The severity of these effects is mainly influenced by the geometry and inclination of the cover glasses.

All cover glasses should be designed in a way that above-mentioned effects are minimized and do not influence the Camera Head's performance. Cover glasses shall not be in contact with the lens. Stray light effects (ghosts, beams, etc.) caused by e.g., cover glasses or stray light covers should not be stronger than the effects agreed in the stray light specification of the Camera Head.

3.1.2.1.4 Entrance Pupil Position

To allow an optimized design of cover glasses, stray light cover and black print Robert Bosch GmbH provides a description of the incoming ray bundle for the Camera Head.
Each ray with a specific object angle has a corresponding convergence point on the optical axis of the camera's lens which is referred to the front lens vertex. By this correlation, each chief ray of the incoming bundle can be described precisely.
In Figure 6 the described correlation is illustrated.
The values will be provided separately in a txt-file on request. The values may change for C-Sample, as they are specific for a lens design.
As it is shown, specifically for fisheye lenses with object angle larger than 90°, the convergence point of a lens also can be virtually in front of the front lens vertex. In this case the distance z is negative.

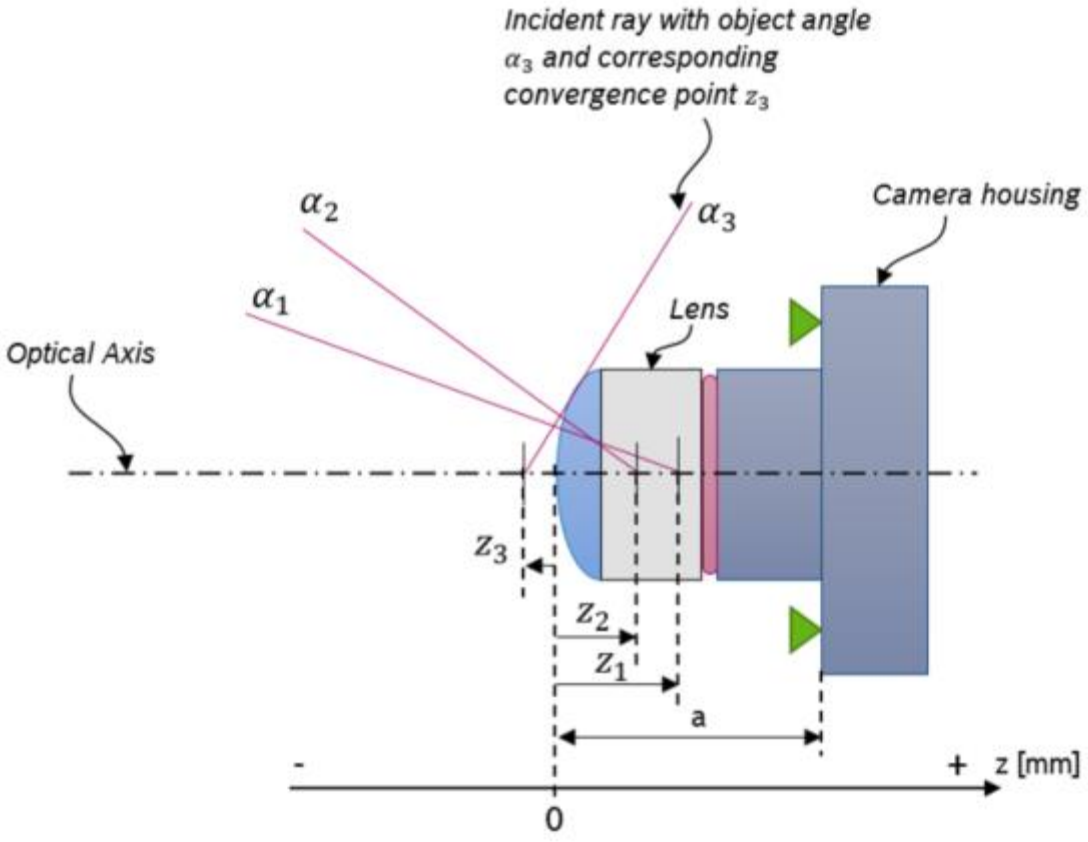


Figure 6: Schematic correlation between incoming ray with object angle α and corresponding convergence point z relative to the lens vertex.

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3.1.2.2 Electrical Interface

3.1.2.2.1 Power Supply

The Camera Head is supplied via Power over Coax (PoC). Data and power are transferred by the same contact pin on the connector and isolating between power and high-speed data signals is done on the camera through a PoC filter.

⚠CAUTION

The supply voltage for the Camera Head on ECU side must be a regulated DC voltage. It cannot be directly supplied by the vehicle board supply. Thereby the power supply shall have a maximum noise as specified in the respective serial link noise specification.

⚠CAUTION

The supply voltage on the ECU must cover voltage drops over the coax cable depending on the actual current consumption of the Camera Head and the actual cable length and cable type.

The main power supply characteristics are shown in Table 2.

Table 2: Electrical interface specification for power supply.

Parameter	Value	Unit	Notes
Nominal Voltage	12	V	Recommended operating voltage at camera connector pin. This value shall be used as design input of the ECU power supply. Voltage drop across the POC of the ECU, coaxial cable and optional inline PCBs (injectors) shall be considered in the system design.
Operating Range	9 - 16	V	Recommended operating voltage at camera connector pin. Voltage drop across the POC of the ECU, coaxial cable and optional inline PCBs (injectors) shall be considered in the system design.
Maximum Voltage	18	V	Maximum voltage to not damage camera. Voltage drop across the POC of the ECU, coaxial cable and optional inline PCBs (injectors) shall be considered in the system design
Voltage Ripple	Compliant to GMSL2 Noise specification.		
Min. Operating Voltage Rise Time	>1	V/ms	

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Typical Power dissipation Lens Heating deactivated	0.8	W	12V camera supply, room temperature, average sample, 60fps
Typical Power dissipation Lens Heating activated	4.7	W	12V camera supply, room temperature, average sample, 60fps
Maximum Power dissipation Lens Heating deactivated	≤1.3	W	9V camera supply, room temperature,, corner sample, 60fps
Maximum Power dissipation Lens Heating activated	≤5	W	9V camera supply, room temperature,, corner sample, 60fps
Overcurrent detection threshold (function state: streaming)	0.4	A	ECU to monitor current consumption of each camera. If the threshold is exceeded, ECU to detect fault condition and turn off power supply of the failed camera.
Overcurrent detection threshold (function state: streaming and lens heating)	0.8	A	ECU to monitor current consumption of each camera. If the threshold is exceeded, ECU to detect fault condition and turn off power supply of the failed camera.
PMIC (Internal)			
PMIC I2C address	0xE0		I2C interface as read-only in case of errors (safety monitoring).
PMIC I2C register address	1	Byte	
PMIC I2C register value	1	Byte	

3.1.2.2.2 Lens Heating Functionality

The Camera Head is equipped with lens heating functionality for use-cases such as removing an ice layer on the front lens.
The power for the lens heating is provided via power over coax.
The power consumption will be increased in case the lens heating function is activated (refer to Table 2).

By default, the lens heating device is switched off and the heating can be activated by the ECU via I2C command.
Controlling of the lens heating includes the following premises / requests to the ECU:

- The ECU has to check both the vehicle temperature sensor and the temperature sensor(s) of the camera head before switching the lens heating on to prevent the system from damage by overheat.
- The ECU has to protect the lens heating module against overheat: The lens heating shall be switched on only if the output value of both temperature sensors is within a pre-defined range. This range will be based on a temperature model of the front lens which has to be developed based on climatic chamber and vehicle validation data.
- The ECU shall have a current limitation and a current monitoring for the Camera Head (including lens heating if applicable).
- Bosch can provide a reference control-loop on request.
- The lens heating shall not be switched on longer that 5 minutes in any case.

General remark on optical performance during active phase of lens heating:
Note that especially during the heat-up phase of the lens heating (i.e. until the lens reaches a thermal equilibrium), a deterioration of the optical performance (reduction of sharpness) can occur.

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⚠WARNING

Based on the measurement data, the ECU has to guarantee that the temperature of the front lens stays below 60°C in situations, when it can be touched (prevention of burning skin; esp. if car is not moving). During situations when the lens cannot be touched (car moving sufficiently fast, e.g. vehicle speed > 6km/h), the front lens could reach higher temperatures.

3.1.2.2.3 Video Data and Control Interface

Video data is transmitted via a MAXIM GMSL2 Link between the serializer in the camera and the deserializer in the ECU. Inside the camera and on the link the video data transmission is based on MIPI-CSI-2 standard.

The serializer is the I2C host in the camera and an I2C slave for the ECU. The serializer forwards all I2C commands from the ECU send by the deserializer towards the internal camera I2C bus.
The serializer provides the clock frequency for the image sensor. For that, the deserializer needs a clock source with a maximum accuracy of +-200ppm. An accuracy of +-50ppm is recommended. Figure 7 illustrates the clock tree in case reference over reverse mode is used.

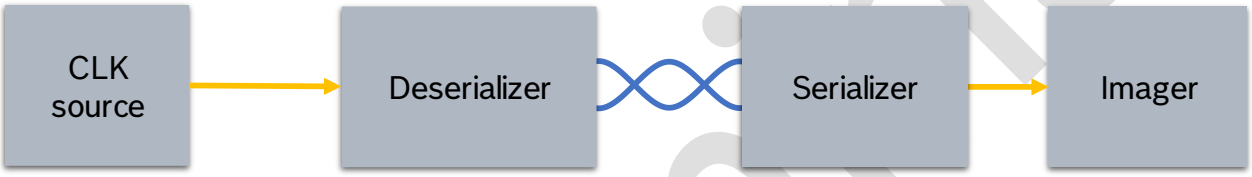


Figure 7: Illustration of clock tree in reference over reverse mode (recommended configuration).

Detailed information on the serializer and its interface characteristics can be found in Table 3.

Table 3: Serializer electrical specification – exemplary for GMSL2.

Parameter	Value	Unit	Notes
Serializer			
Data Interface	GMSL2		
Serializer part number	MAX96717FGTJ/VY+		
Serializer I2C address	0x80 8-bit		
Serializer I2C register address	2	Byte	
Serializer I2C register value	1	Byte	
Serializer clocking mode	reference over reverse		Default configuration for B-sample is Xtal
Serializer video mode	Pixel		Default configuration is Pixel Mode for compatibility with MAX96712.

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			This can be changed via register write if necessary.
Reference clock	25 (preferred value)	MHz	Clock via reference over reverse
Forward-channel data rate	3	Gbps	
Back-channel data rate	187,5	Mbps	
MIPI CSI-2 number of lanes	1 clock + 4 data		
MIPI CSI-2 data rate (from imager)	60fps: 0.620 per lane, 2.48 total	Gbps	
PDB	PMIC Ready:		Pulled High as soon as startup of PMIC Done
MFP0	IMG_RST_N		Enable (high) or disable (low) imager
MFP1	CFG0_FSYNC		Strap during start-up; Frame Sync connection to imager
MFP2	CFG1		Strap during start-up;
MFP3	SER_LH_EN		Enable (high) or disable (low) lens heating
MFP4	SER_CLKOUT		Programmable clock output of the serializer; Used as imager clock input
MFP5	PMIC_ERROR_N		Error input from PMIC; No error (high), error (low)
MFP6	NA		
MFP7	IMG_ERROR_N		Error input from imager; No error (high), error (low)
MFP8	EEPROM_WP		Write protect of EEPROM: high: protected low: write enabled

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3.1.2.2.4 Image Sensor Interface

General

The imager is connected as a slave via I2C to the serializer. The clock signal for the imager is provided by the serializer. To enable the image sensor the reset pin has to be released by the serializer. Any occurring safety or non-safety failure inside the image sensor is linked to the error pin. The error pin is low active.
The imager has a MIPI-CSI-2 interface for transferring image and diagnostic data to the serializer. All four lanes and CLK lane of the MIPI-CSI-2 interface are used.
Detailed information on the imager and its interface characteristics can be found in Table 4 and Table 3.

Table 4: Image Sensor electrical specification.

Parameter	Value	Unit	Notes
Imager			
Imager I2C address	0x20 8-bit		
Imager I2C register address	2	Byte	
Imager I2C register value	2 or 1	Byte	

Reference setting for power-up and basic functionality can be provided upon request.

Synchronization

The synchronization of several image sensors is performed by the ECU. It is possible to synchronize multiple cameras by using the GPIO-Forwarding from the SerDes. For this reason the frame-sync-pin of the image sensor is connected to an output pin of the serializer.

3.1.2.2.5 EEPROM

The EEPROM acts as a slave and is connected via I2C to the serializer.
Detailed information on the EEPROM and its interface characteristics can be found in Table 5.

The EEPROM is configured with the following data during End-of-Line Test:

- General header block describing the EEPROM layout
- Label data
- Intrinsic calibration data
- Security data (not in B-Sample)

Additionally, the EEPROM contains reserved memory space for failure entries and temperature histogram data. These additional data blocks must be written by the central ECU during operation.
Please consider the hints regarding failure entries mentioned in chapter "7 Assessment of Products Returned from the 0-mileage and from the Field".
The data on the EEPROM is stored in big-endian format.
The EEPROM data map (incl. default values after end-of-line test) and details regarding intrinsic calibration can be provided upon request.

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Table 5: EEPROM electrical specification.

Parameter	Value	Unit	Notes
EEPROM			
EEPROM memory size	32k	Bit	
EEPROM I2C address	0xA0,0xB0 8-bit		First address for memory access; Second address for identification page
EEPROM I2C register address	2	Byte	
EEPROM I2C register value	1	Byte	Multiple bytes write is possible

3.1.2.2.6 Security Key Handling Concept

Preliminary Description

Note: Detailed scope and requirements for cyber security are not available yet, which shall be analyzed and aligned between Bosch and the customer.

From C1 sample on the camera will provide a security key for authentication within the ECU. In general, the concept for key provisioning is as follows:

- Image sensor supplier generates the root key pair for the customer.
- Image sensor supplier generates device specific keys and device certificates based on root key pair.
- Image sensor supplier fuses device specific keys and device certificates on each imager.
- Image sensor supplier provides public root key to customer.

3.1.3 Vehicle Integration Interfaces

The Camera Head will be provided to the customer either as the camera head itself or with a pre-mounted bracket. Chapter “3.1.3.1 Mounting Interface” to chapter “3.1.3.4 Poka Yoke and Camera Orientation” refer to camera heads without pre-mounted bracket.

3.1.3.1 Mounting Interface

The Camera Head provides defined reference surfaces for mounting the camera.

For alignment along the x-axis, contact points at the front side of the camera should be chosen within the areas marked green in Figure 8 (a). The holding force shall be applied within the red rectangle-shaped area of the back cover, see Figure 8 (b). The resulting push force shall not exceed 60 N with a max. surface pressure of 30 N/mm2 and shall ensure contact of the front side reference.

For vertical alignment along the z-axis of the Camera Head’s housing, two contact points within the green highlighted rectangular area at the bottom housing surface in Figure 8 (c) shall be set to serve as the interface for the vehicle bracket. The holding force along the z-axis shall be applied within the rectangular support area at the top housing surface in Figure 8 (a).

For horizontal alignment along the y-axis of the Camera Head’s housing, one contact point within the green highlighted rectangular area at the left housing surface in Figure 8 (c) shall be set to serve as the interface for the vehicle bracket.

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The holding force along the y-axis shall be chosen within the red highlighted rectangular support area at the right housing surface in Figure 8 (a).
The maximum push force for alignment along the z- and y-axis is 30N.

The reference surfaces also act as reference for the alignment of the optical path. If other contact points are chosen, the specified FOV and tolerances in pitch, yaw and roll angle cannot be guaranteed.
Exact area of reference and support surfaces may slightly change with C-Sample.

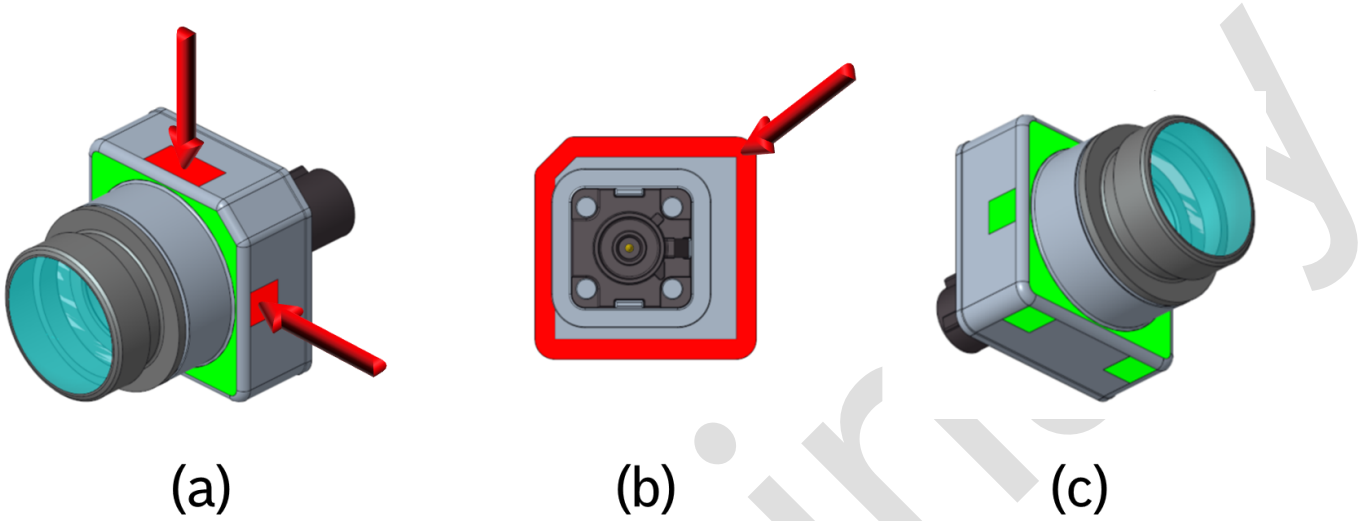


Figure 8: Reference surfaces highlighted in green and areas for application of holding force highlighted in red of Camera Head

3.1.3.2 General recommendations bracket design

The interfaces of the Camera Head including the mechanical reference planes illustrated in Figure 8 are also given in detail in the offer drawing. 3D-CAD data of the Camera Head shows the nominal alignment of the camera, the field of view cone and the bracket interface.

The Camera Head is mounted at defined positions in the vehicle using a dedicated camera head bracket. The bracket design is considered to be in responsibility of the OEM.
The OEM shall coordinate the specific bracket designs for each Camera Head variant and mounting position with the Robert Bosch GmbH. Robert Bosch GmbH can supply a vehicle bracket reference design to the OEM as a starting point and to accelerate the OEM in design phase of customer specific vehicle bracket. Additionally, Robert Bosch GmbH is supporting the OEM in designing the camera head bracket and offers reviews.

The bracket of the Camera Head shall be designed such, that the field of view (FOV) is not blocked by other vehicle components close to the camera.
The bracket must hold the Camera Head in place over lifetime without allowing any displacement of the Camera Head referenced to the vehicle. Considering the specific mounting position of the Camera Head in the vehicle, the fixation using the bracket must prevent self-oscillations that could lead to a system functionality degradation.
The vehicle bracket must be designed for the camera specific weight.
Only the Camera Head's front lens and the area around the front lens should be exposed to direct sunlight.
Furthermore, the bracket may avoid water pooling and water dropping on the Camera Head from components mounted above the Camera Head

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The coax connector engagement force for plugging and unplugging should be considered in the bracket design. Please refer to values given in connector interface specification.

Please consider the hints regarding thermal heat transfer of the vehicle bracket design in the context of maximum ambient operating temperatures described in chapter "4.4 Climate Characteristics".

3.1.3.3 Galvanic Isolation from Vehicle Ground

The Camera Head must be installed into the vehicle with galvanic isolation of the Camera Head's housing and cover to the vehicle chassis ground.
Plastic material is recommended for the bracket design as it supports a galvanic isolation between the camera housing and the vehicle grounding. Preferred plastic material should absorb maximum thermal radiation from the Camera Head.
In case metallic clamping springs will be used, those must be galvanic isolated within the bracket systems against vehicle grounding. To avoid galvanic corrosion the spring material selection should consider the Camera Head housing and cover materials indicated in the offer drawing.

3.1.3.4 Poka Yoke and Camera Orientation

The Camera Head exhibit a chamfer at one edge of the housing (Figure 13) which shall be used as poka yoke coding and ease the installation of the Camera Head in the correct orientation into the vehicle bracket. In chapter "4.8.2.2 Camera Orientation" the default position of the poka yoke chamfer and correct camera orientation are defined.

It is in responsibility of the OEM to design the bracket in a way that different Camera Heads cannot be mixed up in the mounting space.

3.1.3.5 Camera Head with Pre-mounted Bracket

The Robert Bosch GmbH coordinated the specific bracket design. The bracket is mounted at defined positions in the vehicle and the bracket interface design is considered to be in responsibility of the OEM. OEM supplied a vehicle bracket reference design to the Robert Bosch GmbH as a starting point and to accelerate the Robert Bosch GmbH in design phase of customer specific vehicle bracket. Additionally, OEM is supporting the Robert Bosch GmbH in designing the camera head bracket and offers reviews.

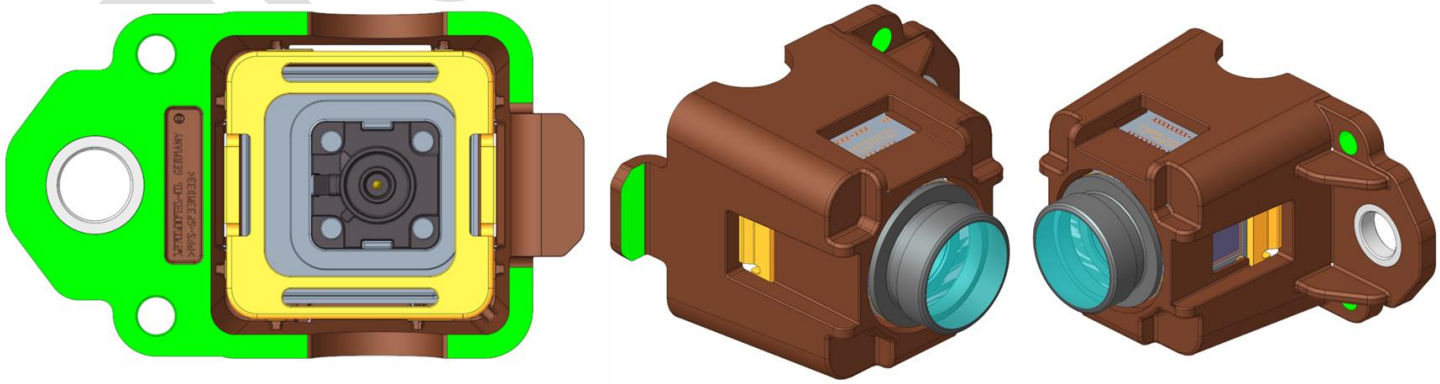


Figure 9: Illustration of camera head with pre-mounted bracket.

Figure 9 illustrates the camera head with pre-mounted bracket. The interfaces of the bracket including the mechanical reference planes are marked in green.

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The bracket provides fixation of the camera head using a spring element. This bracket holds the Camera Head in place over lifetime and avoids displacement of the camera referenced to the vehicle.
The bracket of the Camera Head is designed in a way, that the field of view (FOV) is not blocked by other vehicle components close to the camera.
Only the Camera Head's front lens and the area around the front lens will be exposed to direct sunlight.
Responsibility for screw joint (Tightening torque, fatigue resistance, Safety against untwisting, dimensioning of the screws) is up to OEM.

Volvo specifies the tolerances of bracket reference surface (marked in green) towards camera mechanical axis in Volvo specification drawing.

Volvo is responsible for the vehicle installation interface. Bosch complies with Volvo specification drawing number 78795952 with 4sigma tolerance distribution.

The assembly tightening torque for the bracket is defined as 6 Nm +/- 1 Nm.
The screw material shall be chosen by the customer in a way that contact corrosion is avoided towards the bracket.

3.2 Product interface

⚠CAUTION

Plug connection system according to customer request. Customer is responsible to assure specification conformity and reliability. Robert Bosch GmbH is not liable for defects on the wiring harness plug and any possible consequential defects resulting thereof and is exempt from third party claims. Bosch assures component-related interface conformity in accordance with customer specification.

3.2.1 Mechanical interface of connector plug

The Camera Head has a coded FAKRA connector, which also provides the electrical interface. The FAKRA connector is integrated in the back cover of the Camera Head.
The forces for FAKRA connector plugging and unplugging as given in the connector interface specification are valid (forces to be considered in bracket design).

The FAKRA connectors contains the following characteristics:

- Connector variant: sealed version
- Mechanical coding: A see Figure 10
- Colour coding: A (jet black, RAL 9005)



Figure 10: FAKRA connector coding

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3.2.2 Electrical interface of connector plug

The coax connector contains two contacts with the following configuration:

- Centre pin**:** Power over coax (supply voltage) and data transmission.
- Outer shield**:** GND connection.

3.2.3 Wiring Harness

The cable connector is not in scope of the delivery by Robert Bosch GmbH for series product.
The OEM is responsible for the installation of the Camera Head into the vehicle. This includes the assembly of the Camera Head using a dedicated bracket to the vehicle, the connection of the Camera Head to the vehicle wiring harness and to ensure the tight fixation of the cable close to the Camera Head.
Tight fixation of the wiring harness shall be at <=10 cm distance from the camera plug to relief cable stress and prevent relative movements between the coax connector and the cable plug over lifetime. Tensile or compressive load of the wiring harness to the connector has to be avoided.
Robert Bosch GmbH is testing the robustness of the camera against mechanical shock and vibration using a cable fixation point of 10 cm close to the connector.

The OEM shall avoid a cable installation that promotes water ingress into the Camera Head via the cable e.g., by allowing for stagnant water at the end of the wiring harness.
It is recommended to have a downwards bending of the cable just after the Camera Head's connector.
The Camera Head is designed for a maximum cable length of 15m with 2 cable interconnections (see details depending on cable type and data rate within this chapter).
The following cables have been used as reference:

- Leoni Dacar462, flexible, higher attenuation
- Leoni Dacar302, more rigid, less attenuation

These cables can be replaced by similar cables with the same characteristics (insertion and return loss, impedance, ...): Type RG174 (Dacar462) and type RG223 (Dacar302).
Maximum static bending radius for Dacar462 shall not exceed 10 x diameter or 10 x 2.9mm (29mm). By static bending for Dacar462 Robert Bosch GmbH implies the situation when the total number of cable bending will not exceed 100 times during lifetime of the vehicle.
If Dacar302 cable is used the maximum static bending radius recommended by Robert Bosch GmbH is 20 x diameter or 66mm. By static bending for Dacar302 Robert Bosch GmbH implies the situation when the total number of cable bending will not exceed 10 times during lifetime of the vehicle.
Robert Bosch GmbH recommends to use Dacar302. If Dacar462 is needed due to is flexibility in the installation environment, Robert Bosch GmbH recommends to use a cable combination where Dacar462 will have short lengths (<1m, e.g. for installation in door mirrors) and Dacar302 will have the longer length.

⚠CAUTION

When selecting suitable cables, the OEM must consider the increased currents for lens heating. The higher the resistance of the cable, the higher is the voltage drop between ECU and Camera and the higher is the power loss on the cable.

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4 Technical data with measured variables and measuring conditions

4.1 Functions, function states (modes of operation), functional characteristics and boundary conditions

The camera head's functionality after ISO16750-1 Function State A is as follows:

- Streaming of raw video data and meta data (status information) over LVDS with defined framerate (frames per second). Thereby the camera head itself acts as a slave and has to receive control commands by an external host.
- Specified power consumption
- Internal CRCs and error flags for safety monitoring
- Optical characteristics (refer to chapter "3.1.2.1.1 General Optical Characteristics") are within the limits of the specification.
- Intrinsic calibration parameters stored on the EEPROM of the camera head during production.

4.2 Mechanical characteristics

Mounting interfaces:

- Please refer to chapter "3.1.3.1 Mounting Interface".

Forces:

- Please refer to mounting interfaces (see above) and to chapter "4.8.2.3 Forces on Camera Head".

Sealing:

- Please consider Figure 11
- Front sealing (camera mounted in vehicle bracket): tightness classes IP6K6K, IP6K7, IP6K9K.
- Back sealing (cable plug mounted and connector interface supports the following sealing classes): tightness class IP6K4K, IP6K7, IP6K9K* (*direct water beam to backside to be prevented by mounting position in vehicle)

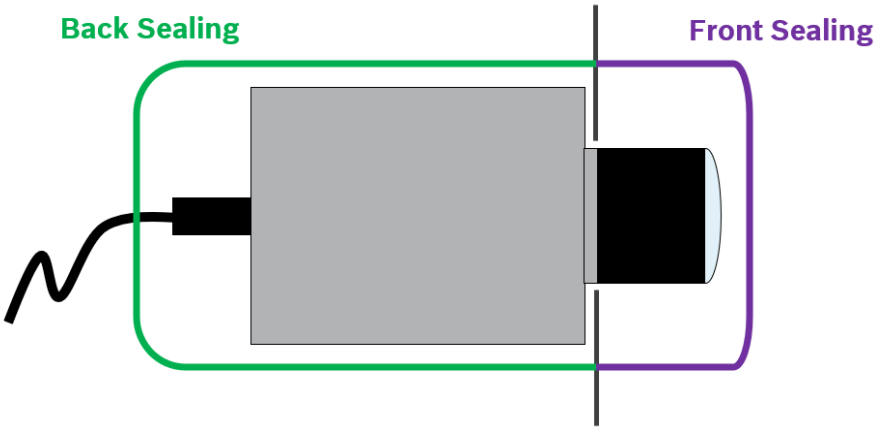


Figure 11: Sealing of camera head

Vibration and mechanical shock (max. permissible loads):

- Random vibration: 8h for each direction, effective acceleration: 30.8 m/s², T_{min} and T_{max}
- Mechanical shock: effective acceleration 300 m/s²

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4.3 Electrical characteristics

Power Supply:

- Please consider the values and hints given in chapter "3.1.3.2 Electrical Interface".

EMC:

- Regarding maximum limits please refer to EMC test plan in chapter "6.1.3.3 EMC / ESD Validation".

ESD

- Please consider the hints in chapter "4.8.2.5 ESD Protection".
- Regarding maximum limits please refer to ESD test plan in chapter "6.1.3.3 EMC / ESD Validation".

4.4 Climatic characteristics

Operating ambient temperature:

Operating ambient temperature:

- The camera head is targeted to be fully functional from -40°C ... +85°C camera ambient temperature.
- A realistic convection and an appropriate heat transfer from the camera to the surrounding (via the vehicle bracket) has to be considered.
- The temperature sensor of the image sensor shall be monitored in order to ensure that the junction temperature is below the max. allowed value.

Transport conditions:

- Please refer to chapter "4.8.1.3 Transport Condition".

Storage conditions:

- Please refer to chapter "4.8.4 Storage".

4.5 Chemical characteristics

The camera head will be tested for the specific agents based on ISO16750-5 for customer C-sample.

4.6 Acoustic characteristics

Not applicable, the camera head has no acoustic characteristics

4.7 Lifetime

With respect to the use and usage conditions described in this TCD, the life of the product is designed for the parameters stipulated in the table below.

The product lifetime is achieved as soon as one of the stipulated parameters is reached (independently of the remaining parameters):

Lifetime (Product)	15 years
Milage	2.000.000 km
power-on time	30.000 h

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This lifetime statement must be differentiated from the applicable warranty period which expires independently from the duration of this lifetime statement.
Bosch trusts that the OEM will assess if the Bosch-product's lifetime specified in this TCD is in line with the relevant legal lifetime requirements applicable to the OEM's system in the specific target country and whether any particular measures need to be taken in order to comply with relevant legal lifetime requirements (e.g. if the Bosch-product needs to be replaced after the Bosch-product's lifetime period).

4.8 Transport, assembly, start and end of operation, storage

Please pay special attention to the safety and warning notes!

4.8.1 Transport

4.8.1.1 Packaging of Camera Head in RB plant

The camera heads are packaged in an ESD tray. The number of cameras within one blister will be defined during project development depending on camera size. Several trays can be stacked on each other and form a stack. The full stack is packaged by an ESD bag for further packaging in a cardboard box. Figure 12 illustrates exemplarily the tray and stack for a camera head of housing size 23x23 mm.





Figure 12: ESD tray for camera heads.

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4.8.1.2 Packaging and shipping of preassembled components from Tier1 to OEM

This chapter highlights the packaging recommendations of pre-assembled components, where the Tier1 mounts a Camera Head into the component, e.g. mirrors, and ships the component to the OEM for final vehicle assembly. Robert Bosch GmbH recommends considering the following topics related to Tier1 packing:

⚠WARNING

Consider ESD protection during handling and packing e.g. mirror holds electronic components (Camera Head)
Protect the front lens of Camera Head from damage during packing and transport
Prevent the Camera Head's lens surface from abrasion during transport due to direct contact to packaging material
Protect the Coax connector from damage during packing and transport

4.8.1.3 Transport conditions

During transport the environmental conditions can exceed the specified long time storage conditions (referring to conditions listed in chapter "4.8.4 Storage" on Page) for a short time period, e.g. during transportation via plane in an unopened original package.
In any case, the following conditions shall hold during transportation:

- The minimum temperature during transportation is −40°C.
- The maximum temperature during transportation may not be above +95°C
- The relative humidity must not exceed a value of 80%

⚠WARNING

The Camera Heads must be carried in their original package. Throwing, mechanical shocks or heavy strokes of the packaging has to be avoided at all events.

⚠WARNING

The risk of corrosion by transportation in corrosive environments (e.g. intrusion of salt water) must be avoided. In terms of sea freight, a suitable special packaging is needed for such kind of shipping (e.g. Dry-Pack). For B-sample such a package is not yet available. Sea freight in Robert Bosch GmbH standard packaging is not permissible.

4.8.2 Assembly

The interfaces of the Camera Head including the mechanical reference surfaces are described in chapter "3.1.3.1 Mounting Interface".
Recommendations regarding vehicle integration and bracket design are given in chapter "3.1.3.2 General recommendations bracket design".
The interfaces of the camera with pre-assembled bracket are described in chapter “3.1.3.5 Camera Head with Pre-mounted Bracket”.

4.8.2.1 Unpacking and Typical Assembly

The Camera Head is shipped in an appropriate tray, in which the lens is protected to be touched by other components. After the Camera Head is taken out of the tray utmost care must be taken in handling it during the mounting and fixation process in the OEM assembly line.

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⚠WARNING

Damage, pollution or scratches especially at the front lens of the Camera Head or damage of the connector during assembly must be avoided. The front lens shall not be touched during assembly process.

Bosch recommends that operators are wearing appropriate gloves during installation of the Camera Head into the vehicle and to test the main lens after installation visually for not being damaged or contaminated.

Recommended Unpacking Procedure:

- Obey the ESD measures
- Open the ESD bag to get the trays out of it
- Take the Camera Head out of the tray

Typical Assembly:

- Refer to hints in following sections and recommendations given in chapter "3.1.3 Vehicle Integration Interfaces"

4.8.2.2 Camera Orientation

The default orientation for the Camera Head is shown in Figure 13

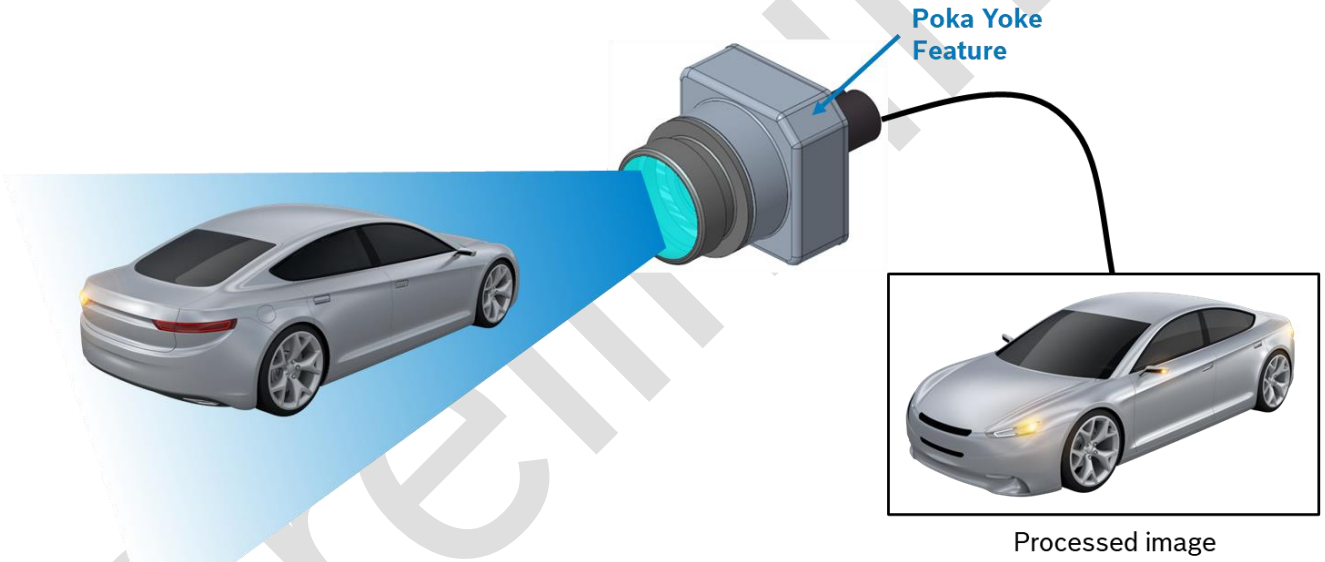


Figure 13: Default orientation of Camera Head.

Each Camera Head installation position and orientation shall be agreed between Robert Bosch GmbH and OEM.

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4.8.2.3 Forces on Camera Head

4.8.2.3.1 Clamping Forces

WARNING

Prevent a clamping of the Camera Head's housing beside the reference and support areas in radial direction during installation in the assembly line and over lifetime.

The maximum clamping forces are given in chapter "3.1.3.1 Mounting Interface".

4.8.2.3.2 Lateral Forces to Lens Barrel

WARNING

Lateral forces/shocks to the Camera Head's main lens barrel during installation and at the mounting position over lifetime shall be avoided.

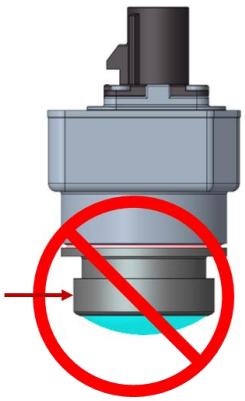


Figure 14: Lateral forces on Camera Head's main lens (red arrow) that shall be avoided.

4.8.2.3.3 Forces on Camera Head Housing

WARNING

Prevent all forces on the Camera Head housing during installation in the assembly line and over lifetime which are not included in the above described forces for mounting purpose.

4.8.2.4 Dropped Devices

WARNING

Dropping of the Camera Head during installation process shall be avoided. Accidentally dropped devices must be discarded due to potential damage of the main lens, the connector or the housing to bracket interface.

4.8.2.5 ESD Protection

WARNING

The Camera Head is an electronic device and sensitive to electrical discharge. Therefore, ESD standard protection according to IEC 61340-5-1 and IEC 61340-5-2 must be obeyed at all times during unpacking, handling, installation

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and transportation. E.g. Touching/ interfacing the Camera Head's coax connector with others than the foreseen coax cable connector, e.g. tools, blank wires, etc. is not allowed at any time.

4.8.2.6 Coax Connector Assembly Hints

The FAKRA Connector is integrated in the back cover of the Camera Head. Details regarding coding and colour marking are described in the chapter "3.2.2 Coax connector, coding and pin assignment".
At least 4 mating cycles are left for the coax connector after delivery of cameras to the OEM.
The Camera Head's FAKRA connector plastic tube surface is part of the connector sealing concept and may not be damaged during installation.
When the cable connector is mounted, the connector must snap audibly into the port of the Camera Head.

⚠WARNING

Camera Heads with a connector that has been damaged during installation must be discarded.

Specified IP classes (see offer drawing) of the Camera Head are met only with plugged connector of the wiring harness.

⚠WARNING

The coax connector interface shall be protected against environmental loads as long as the connector of the Camera Head is unplugged, as for example dust could degrade the sealing capability of the connector when mated and water on the electrical contacts of the connector could lead to corrosion and hence to reduced signal integrity.

Appropriate measures during storage and assembly of the Camera Head have to be considered (e.g. covering the connector) to avoid contamination or water ingress into the connector.

4.8.2.7 Continuity Test

For third party component suppliers, where the Camera Head has to be installed prior to final vehicle assembly at OEM, we recommend the following test setup to check whether the cable was assembled correctly.

After the Camera Head is assembled to the bracket and the cable connector was plugged:

- Connect the Coax cable connector to a dedicated test box with an according coax interface
- Test box to apply a nominal voltage to the Camera Head of 12 V +/- 5% and current limiting capability
- Use a test box having an integrated current protection
- The test box checks after 1 s whether the current supplied to the camera is in specified range
- The test box ramps down voltage after test automatically

Specifications:

- Camera Head test voltage: 12 V +/- 5%
- Camera Head voltage ramp up slope: >1V/ms
- Test box current limit: < 1 A
- Cable length: 2m < length < 10m
- Camera Head current consumption range: 10 mA < test limit < 120 mA *(Note: Upper limit is used to detect a potential short)*

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4.8.3 Start and End of Operation

4.8.3.1 Start of Operation

The start-up sequence of the camera head will be provided during project development phase, as the exact sequence depends on the overall system architecture and the safety & security concept.

4.8.3.2 End of Operation

⚠WARNING

Following instructions have to be considered, if the Camera Head has to be removed from the installation position:

- The Camera Head must neither be damaged nor deformed at the housing. Around the area of the lenses every kind of force effects or contacts are strictly prohibited.
- Before demounting the Camera Head, suitable ESD protection procedure must be taken. Moreover, the ESD protection standard DIN EN 61340- 5-1 must be observed.
- The lenses must not be contaminated. In order to protect it after demounting, the lenses must be sheltered.
- Before demounting the Camera Head, the wiring harness connector at the back of the housing has to be removed carefully, by unlocking the connector position assurance (CPA) slider.
- Please be aware that during detaching, the connector and pins must not be damage.

In the event of returning a single Camera Head to BOSCH, the device must be packed adequately as described in Chapter "Transport" and the shipping storage conditions must be maintained as mentioned.

4.8.4 Storage

⚠WARNING

The Camera Head must be stored in the unopened original packaging. As soon as the Camera Head is unpacked, it must be installed in the vehicle. Otherwise, the storage state is not defined.

The following conditions must be considered for storage:

- The minimum storage temperature is +10°C
- The maximum storage temperature is +40°C.
- The relative humidity shall have an arithmetic mean value of <=60%
- The maximum storage period for aftermarket support is 5 years.

During transportation, the conditions listed in chapter "4.8.1.3 Transport Condition" must be considered.

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5 Series-accompanying tests

Product audits are performed according to RB internal Quality regulations. The audit tests in series production are performed with a representative unit of the specific product family.

Series-accompanying tests will be defined during project development.

Test designation:	[Test designation, purpose] →TBD by customer project
Goal:	[Goal] →TBD by customer project
Test profile:	[Test profile] →TBD by customer project
Test environment:	[Test environment] →TBD by customer project
Duration:	[Duration] →TBD by customer project
Criteria:	[Criteria] →TBD by customer project

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6 Testing

6.1 Testing by Bosch

Camera head validation cover hardware tests only. It does not include validation on system level. Additionally, the camera head does not contain any software which can be validated.
Besides default validation tests (e.g. AEC-Q100) at component supplier, the tests described in the following sections are performed on camera head level.

6.1.1 Climatic-dynamic Validation

Test designation: Climatic dynamic validation

Goal: Verification of robustness against external loads

Test profile:

Load	Test	Test reference
Temperature Loads	Temperature step test	ISO 16750-4
	Temperature cycling tests	
	Rapid change of temperature	
	Ice water shock test	
	Parameter test	ISO 16750-1
Physical Loads	Dust test	ISO 16750-4
Water Loads	IPX6K, IPX7, IPX9K	ISO 20653
Chemical Loads	Leakage and function test	ISO 16750-4
	Corrosion test	
	Damp heat cyclic test	
	Composite temperature / humidity cyclic test	
	Lens heating test	Bosch
Mechanical Loads	Vibration (over Temperature)	ISO 16750-3
	Mechanical shock	

Test environment: Refer to details in validation test plan

Duration: Refer to details in validation test plan

Criteria: Fully functional, refer to chapter 4.1

6.1.2 Electrical Validation

Test designation: Electrical validation

Goal: Check robustness of electrical performance

Test profile: Test profile will be defined within customer project

Test environment: Refer to details in validation test plan

Duration: Refer to details in validation test plan

Criteria: Electrically functional, refer to chapter 4.1

6.1.3 EMC / ESD Validation

Test designation: EMC / ESD validation

Goal: Check robustness regarding EMC / ESD

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Test profile:	Test profile will be defined within customer project
Test environment:	Refer to details in validation test plan
Duration:	Refer to details in validation test plan
Criteria:	Electrically functional, refer to chapter 4.1

6.2 Testing by the customer

The customer is responsible for the functional system safety validation and therefore responsible for verifying and demonstrating that the product is fit for its intended use and is responsible for the correct calibration of the product.

The customer has the system responsibility and needs to validate the complete system. Bosch is responsible for the System of Interest only. Findings pointing to any component within the SOI will be reported via a dedicated defect management system.

6.2.1 Testing by the customer

Test designation	[Test designation, purpose] →TBD by customer project
Goal	[Goal] →TBD by customer project
Test profile	[Test profile] →TBD by customer project
Test environment	[Test environment] →TBD by customer project
Duration	[Duration] →TBD by customer project
Criteria	[Criteria] →TBD by customer project

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7 Assessment of products returned from 0-mileage and from the field

For claims please consider the acceptance criteria (functional characteristics of the product) described in chapter 4.1. Acceptance criteria not valid for parts that were damaged due to mishandling or were potentially damaged due to inadequate packaging during shipping.
As described in chapter "3.1.2.2 Electrical Interface" the EEPROM contains a reserved memory space area to store failure codes (diagnostic trouble codes, DTCs) which could occur during operation of the camera. The DTCs have to be written by the host (central ECU) to the camera head.

⚠CAUTION

The following points have to be considered by the customer:

- All DTCs which affect the camera head and which could lead to a potential return of the camera head have to be stored on the EEPROM of the camera head.
- The customer has to provide a decoding list for the above mentioned DTCs

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8 Appendices and references

Table 6: References

Nr	Name	Document Nr.	Description
1	ISO 20860-1	-	Road vehicles - 50 Ω impedance radio frequency connection system interface Part 1: Dimensions and electrical requirements

Table 7: Definitions, Abbreviations and Symbols

Abbreviation	Description
ADAS	Advanced Driver Assistance System
CPA	Connector Position Assurance
DC	Direct Current
DIA	Development Interface Agreement
DTC	Diagnostic Trouble Code
ECU	Electronic Control Unit
ESD	Electrostatic Discharge
FOV	Field of View
GMSL	Gigabit Multimedia Serial Link
HW	Hardware
I2C	Inter-Integrated Circuit
MP	Mega Pixel
OEM	Original Equipment Manufacturer
PoC	Power over Coax
Rth	thermal resistance
SOTIF	Safety Of The Intended Functionality
SW	Software
TCD	Technical Customer Documentation
TSC	Technical Safety Concept

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9 History

Release	Date	Edited by	Release description
0.1	2023-01-27	Markus Biró (XC-AC/PAC)	Initial version
0.2	2023-05-13	Markus Biró (XC-AC/PAC)	TCD for RFP2 phase (SP-Version 1.0)
2.0	2025-04-16	Annika Wilms (XC-AC/ECT5)	Complete rework due to template update and reference to latest released PF-TCD

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10 Contact person

Contact person:

- XC-AC/ECT4, Ziegler, Marco

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ADAS Components

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